A five-mile tether, coiled and bifilar, in orbit

Does coiling a tether, or winding it as Tesla's bifilar coil, make it an energy source? — CLAUDE.md §3e, §3f

kronspice simulation report · 2026-11-07 · orbit 7.7 km/s, B = 30 μT, 816 W solar array (declared)

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Summary

A 5-mile (8,046.72 m) copper tether of 2 mm wire (43.0 Ω) in low Earth orbit is compared in three windings: straight (the ordinary electrodynamic tether), an ordinary single-layer coil 2 m across with 1,281 turns, and the same coil wound as Tesla's bifilar coil (US 512,340), in which each turn lies next to the turn half a coil away. The question is whether "multiplication between windings" makes the coil an energy source. Energy comes only from declared sources: the Sun, and Earth's magnetic and electric fields.

- **Coiling throws the field's voltage away:** the straight tether gives 1858.792 V, the ordinary coil 0.888 V and the bifilar coil 0.889 V, which is $E' = v \times B = 0.231$ V/m times the coil's own 3.84 m length.
- **The Sun can drive the coil at resonance:** 816 W at 565.7 Hz (Q 112.5) ring 265.1 V up to 29,784 V peak, ×112.
- **Bifilar moves the energy between the turns, not into the books:** 25.78 J of the 25.783 J stored sits between the bifilar turns against 6.283×10^{-5} J for the ordinary coil, the same total.
- **Every joule is from the Sun:** in 309.92 J = heat 284.13 J + stored 25.781 J (integration error 0.0064 J); a matched load takes half the input (406.9 W) and never more.
- **The bifilar turn gap is overstressed:** 14.9 MV/m between turns against the declared 10 MV/m limit.
- **No multipactor at 566 Hz** (growth per impact 0.063–0.071); avalanche only at 1 GHz and 3 GHz, where the 8 km wire is many wavelengths long.

What is being tested

The idea under test is that winding a tether into a coil, or as a bifilar coil whose adjacent turns differ by half the coil voltage, multiplies what it can harvest. **Standard prediction:** in a uniform field the EMF between a wire's ends is $(v \times B) \cdot (r_{end} - r_{start})$ whatever the path, so coiling cancels it; a resonant coil multiplies voltage by Q but stores only what its drive delivers, and the bifilar winding changes where the energy is stored, not how much. This example computes all of it with the energy ledger on. The multipactor check between turns is the one built for double_hull (§3i), and the burst-speed limit is the rotor rule of §3l.

Declared inputs

input	value	note
wire	5 international miles of 2 mm copper	8,049 m wound (1,281 whole turns), 43.0 Ω DC
orbit speed	7,700 m/s	relative to the co-rotating field
Earth's field B	30 μT	order of magnitude in LEO; a real orbit takes B from IGRF
solar array	2 m² at 30 %, TSI 1,360.8 W/m²	Kopp & Lean 2011; 816 W to the drive
coil	2 m across, pitch 3 mm	1 mm bare gap between turns, vacuum (ε = 1); 3.84 m long
gap limit	10 MV/m	to be replaced by the chosen insulation's data-sheet value

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The model

- **Windings** (em.Coil, the lumped first approximation of §3e): Wheeler inductance, 1.362 H for either winding (the two strands of a series bifilar coil carry the same current the same way); the two-wire capacitance between adjacent turns; Terman's skin-effect resistance; and the patent's turn voltage, V/N ordinary and V/2 bifilar. Referred to the terminals, the turn-to-turn self-capacitance is $(N-1)\cdot C_{tt}/N^2$ for the ordinary coil (1.42×10^{-13} F) and $(N-1)\cdot C_{tt}/4$ for the bifilar coil (5.81×10^{-8} F). Their ratio is Tesla's $N^2/4$, 410,240 for 1,281 turns (the patent's 250,000 at 1,000 turns).
- **Earth's field.** The motional EMF $\int(v\times B)\cdot dl$ is integrated along the actual wire path with em.MotionalEMF (the Lorentz force by grade selection, §3f): a straight line, a helix of 36 points per turn, and for the bifilar coil two interleaved strands with the return lead. A fourth path folds the straight tether in half, out and back 1 cm apart. In the craft's frame the same coupling is the electric field $E' = v\times B$, not a second source.
- **The Sun.** A sine source on the light channel in series with the wire resistance, the coil inductance and the capacitance across the coil (the bifilar coil's own, or a tank capacitor of the same 5.81×10^{-8} F for the ordinary coil), with an optional series load. The amplitude $\sqrt{(2(R_{wire}+R_{load})\cdot P_{array})}$ makes the loop draw the array's whole 816 W at resonance. Each run is a kronspice transient at 200 points per period for seven ring-up time constants ($\tau = 2L/R$); peaks and mean powers are taken over the last 20 cycles.
- **Turn gap.** Voltage, field and stored energy between neighbouring turns; em.Multipactor over every amplitude the gap sees during ring-up, with measured as-received copper yield (Baglin et al. 2002), keeping the worst growth.

Results

1. Earth's field: the EMF along the real wire path

winding	EMF	reason
straight 5-mile tether	1858.792 V	$E' \times 8$ km
coiled (ordinary)	0.888 V	$E' \times$ the coil's 3.84 m length: the turns cancel
coiled (Tesla bifilar)	0.889 V	both strands run the same way through the field
straight, folded out and back	1.14×10^{-13} V	out and back cancel

The dipole field's gradient across the orbit adds at most $N\cdot A\cdot v\cdot 3B/r \approx 0.00041$ V, and all of it is quasi-DC: the field seen by the craft changes over an orbit (~90 min), far below any LC resonance. A coil spun at 1 rev/s in the field gives 0.759 V peak at 1 Hz, paid for by the spin, which slows.

2. The Sun drives the coil at resonance

L 1.362 H, C 5.81×10^{-8} F, f 565.7 Hz, wire R 43.0 Ω; the wire is 0.015 λ, so the lumped model holds (valid below 0.1).

	ordinary coil + tank capacitor	Tesla bifilar	bifilar, matched load
Q	112.5	112.5	56.2
drive amplitude	265.1 V	265.1 V	374.9 V
coil voltage (peak)	29,784 V (×112)	29,784 V (×112)	21,064 V (×56)
coil current	6.15 A	6.15 A	4.35 A
between neighbouring turns	23.3 V, 2.33×10^{4} V/m	14,892.2 V, 1.49×10^{7} V/m	10,532.1 V, 1.05×10^{7} V/m
energy between turns / total stored	6.283×10^{-5} J / 25.783 J	25.78 J / 25.783 J	12.89 J / 12.895 J
multipactor growth per impact (worst)	0.063	0.070	0.070
steady state	Sun in 815.2 W = heat 814.2 W	Sun in 815.2 W = heat 814.2 W	Sun in 815.2 W = heat 406.9 W + load 406.9 W
whole run: in = heat + stored (+ error)	309.92 J = 284.13 + 25.781 J (+0.0064 J)	same	155.13 J = 142.23 + 12.894 J (+0.0032 J)

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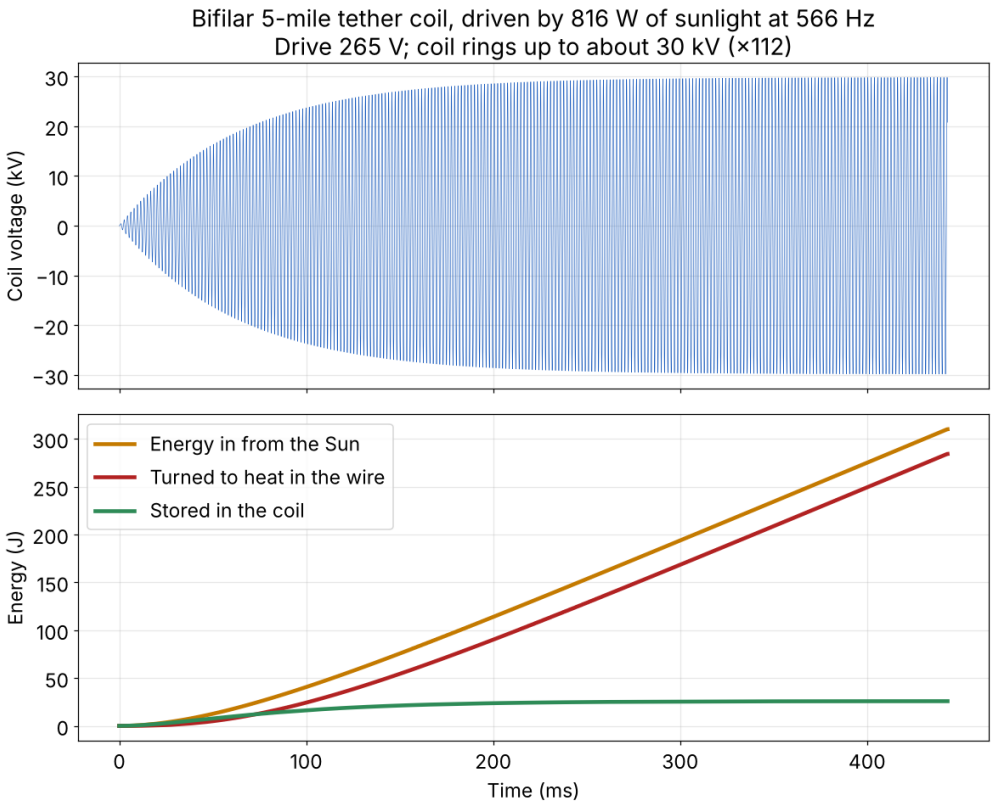


Figure 1. The bifilar coil driven by 816 W of sunlight at 566 Hz. Top: coil voltage ringing up from a 265 V drive to about 30 kV (×112). Bottom: energy in from the Sun, energy turned to heat in the wire, and energy stored in the coil; the stored energy levels off at about 26 J.

Multiplication is real for voltage (by Q), and the bifilar winding puts the whole stored energy between its turns, $N^2/4$ times the ordinary coil's turn-to-turn energy, with the same total. All of it comes from the Sun's array; the matched load takes half the input and never more. The bifilar turn field exceeds the declared 10 MV/m gap limit, so it needs insulation or wider spacing.

3. Driving the resonance from Earth's field instead

The coil would have to spin at 566 rev/s: rim speed 3554 m/s against 1881 m/s, the burst speed of T1000G carbon fibre (the strongest rotor material in spacetime/). It would fly apart.

4. Where electrons would multiply in the 1 mm vacuum gap

frequency	worst growth per impact	at amplitude	verdict
566 Hz	0.071	480 V	no avalanche
1 MHz	0.067	2000 V	no avalanche
100 MHz	0.079	2000 V	no avalanche
1 GHz	1.657	120 V	avalanche
3 GHz	2.029	480 V	avalanche

Worst amplitude scanned up to 2 kV. At gigahertz the electrons take their energy from the drive, and the 8 km wire is many wavelengths long, so it is not a coil resonance.

Ledger and validation

- TestMotionalEMF: straight = vBL to 10^{-12}; each coil = vB × its axial length; folded = 0.
- TestResonance: step-up = Q within 1 %; Sun in = array power = heat within 1 %; ledger closes to 10^{-3}; turn-energy ratio = $N^2/4$ to 10^{-9} with equal totals; matched load takes half the input.
- TestRuns: the whole report runs.
- Whole-run integration error 0.0064 J of 309.92 J (ordinary and bifilar) and 0.0032 J of 155.13 J (matched load).

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- The export sheets carry only the MNA column: automatic tearing finds no valid cut of this single loop (every torn piece is singular, Aitchison 1983 Lemma 2), which the sheet reports instead of solving.

Limits of the model

- B = 30 μT is an order of magnitude; a real orbit takes B from IGRF.
- The coil is the lumped first approximation of §3e (turn-to-turn capacitance model, Wheeler inductance); the distributed and helical-line models are not used here.
- The gap limit (10 MV/m) is a placeholder for the chosen insulation's data-sheet value.

Reproducing the results

```
go run ./examples/coiled_tether
go test ./examples/coiled_tether
KRONSPICE_EXPORT=1 go test ./examples/coiled_tether -run TestExport   # netlists, schematics, results sheets
python examples/coiled_tether/viz/make_report.py examples/coiled_tether/coiled_tether_report.pdf
```

Sources: Kopp & Lean, Geophys. Res. Lett. 38, L01706 (2011); Tesla, US 512,340 (1894), p. 2, ll. 3–7; Wheeler, Proc. IRE 16, 1398 (1928); Terman, Radio Engineers' Handbook (1943); Ramo, Whinnery & Van Duzer, Fields and Waves, ch. 4; Joot, Geometric Algebra for Electrical Engineers, Thm 3.16; Baglin et al., LHC Project Report 472 (2002); Hatch & Williams (1954); Vaughan, IEEE Trans. Electron Devices 35, 1172 (1988); Toray T1000G data sheet (2024); Aitchison, Quart. Appl. Math. 41, 265 (1983).

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